Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/10/2024
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Otolith and Posture Evaluation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	04/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4		
Key Personnel Changes/Previous PI:	Per the PI, Michael Schubert was added as a Co-Inverstigator to this project (Ed., 1/11/24).		
COI Name (Institution):	Schubert, Michael (Johns Hopkins University School of Medicine)		
Grant/Contract No.:	NNX16AO69A-EXP0001		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	EXPAND Project This project will provide information on binocular alignment as a measure of otolith asymmetry – more specifically as a measure of central neural compensation for this asymmetry. This low-level sensorimotor response has several attractive features: • It is easily and rapidly measured with minimal equipment. • It provides complementary information to posture and locomotion tests. • It has been validated in studies in vestibular patients and in parabolic flight.		

	• It can provide an alternate means to assess underlying neurovestibular function which contributes to many sensorimotor responses. We propose here the use of a simple perceptual test to assess binocular alignment, known as vertical alignment nulling and torsional alignment nulling (VAN/TAN), and associated tests of posture and balance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Motion sickness immediately on return to earth remains a problem, with 100% of interviewed astronauts reporting symptoms. This project revisits and updates extensive and promising previous work (mostly by Markham and Diamond) on prediction of SMS susceptibility. Prediction of SMS (and postflight postural instability) would be of great value in setting crew expectations, suggesting crew roles, and eventually for testing and individualized targeting of countermeasures. In addition, for commercial spaceflight, the ability to mitigate SMS would greatly help this industry in expanding its customer base. A version of the apparatus and procedures developed and refined for this project is in use in the vestibular clinic at Johns Hopkins (co-investigator Michael Schubert) and is being further developed for assessment of military personnel with traumatic injuries
Task Progress:	We use a simple perceptual test to assess binocular alignment, known as vertical alignment nulling and torsional alignment nulling (VAN/TAN), and associated tests of posture and balance. In the four subjects of the Inspiration 4 crew, we found the following: • In vertical misalignment (VAN), there is a difference (retrospectively) between subjects who experienced space motion sickness (SMS) and those who did not. When a subject has consistent misalignment before and after flight, and that misalignment is different after flight than before, this is associated with lack of SMS. This is presumably because postflight testing reflects the inflight adaptive state, and that 0g state should in general be different from the 1g state. • TAN (torsional skew) data did not show an obvious pattern. • We measured body sway during quiet standing (eyes closed, subject holding tablet computer to chest). There were no obvious differences in sway magnitude or frequency content between subjects with and without SMS. A more sophisticated analysis (Detrended Fluctuation Analysis) examines fluctuations (variance) around a linear trend in the sway data, at different time scales. Many physiological processes follow a power law, where the magnitude of the fluctuations (F) is related to the duration of the time window (T) over which the fluctuations are measured: F(T) ~ (T)^a. This demonstrates that there are temporal correlations within the data, over many time scales, which are organized and not haphazard. Those subjects who experienced SMS show a deviation from power-law scaling. These results hold promise for using ocular alignment to predict SMS susceptibility and alterations in postural control.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Significant Media Coverage	Chang K. "Who gets sick in space? Orbital tourists may offer better clues." New York Times, September 16, 2021. https://www.nytimes.com/2021/09/16/science/inspiration4-medical-research.html , Sep-2021